

MATH 3115 MSC & SED COMPLEX ANALYSIS

Lecture 2: Polar coordinates in complex numbers

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Lesson Outcome

By the end of this lesson you should be able to:

- Write complex numbers in polar coordinates.
- Write the complex numbers.
- Product of complex numbers in polar form
- Divide Complex numbers

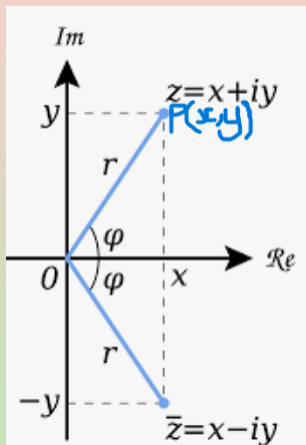
INTRODUCTION

- Complex numbers are a powerful mathematical tool that extends our understanding of the physical world.
- They enable us to solve problems that would be difficult or impossible to address with only real numbers.
- They have broad applications in both theoretical and practical domains.
- In this lecture we discuss about the polar coordinates and solve the product and quotient of complex numbers

Complex numbers in Polar form

- Each complex number is uniquely determined by two polar coordinates r and θ
- where r is the non negative length of the segment OP joining the point P to the origin.
- while θ is the angle from the x axis to this line segment.

Complex numbers in Polar form CONT'D



- The polar coordinates (r, θ) of $z = x + i y$ can be determined by using trigonometry SOH CAH TOA

Complex numbers in Polar form CONT'D

- $\cos \theta = \frac{x}{r} \Rightarrow x = r \cos \theta$
- $\sin \theta = y/r \Rightarrow y = r \sin \theta$
- $\tan \theta = y/x, x \neq 0 \Rightarrow \theta = \tan^{-1}(y/x)$
- $r^2 = x^2 + y^2 \Rightarrow r = \sqrt{x^2 + y^2}$
- The number r is the modulus of z .
- The angle θ is called the argument of z .
- We write $\theta = \arg z = \tan^{-1}(y/x)$.
- With these relation, we can write a complex number $z = x + iy$ in polar form.

Polar coordinate CONT'D

Definition 1

If $z = x + iy$, then $z = r \cos \theta + ir \sin \theta$. Hence, we have

$$z = r(\cos \theta + i \sin \theta)$$

Example 2

Write the complex number $z = \sqrt{3} + i$ in Polar form.

Polar coordinate CONT'D

Solution 3

This complex number $z = \sqrt{3} + i$ is in the first quadrant so $0 < \theta < \pi/2$

$$x = \sqrt{3}, y = 1, \quad \theta = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \Rightarrow \theta = \pi/6$$

$$r = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{3 + 1} = \sqrt{4}$$

$$r = 2$$

$$z = \sqrt{3} + i = 2(\cos \pi/6 + i \sin \pi/6)$$

Polar Coordinate CONT'D

Example 4

write $(1 + \sqrt{3})^2$ in polar form.

Polar Coordinate CONT'D

Solution 5

$$\begin{aligned}(\sqrt{3} + i)^2 &= (\sqrt{3} + i)(\sqrt{3} + i) = \sqrt{9} + \sqrt{3}i + \sqrt{3}i + i^2 \\&= 3 + 2\sqrt{3}i - 1 = 2 + 2\sqrt{3}i\end{aligned}$$

$$= r = |z| = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = \sqrt{16} = 4$$

$$\theta = \tan^{-1} \left(\frac{2\sqrt{3}}{2} \right) = \tan^{-1}(\sqrt{3}) = \pi/3$$

$$\theta = \frac{\pi}{3}$$

$$(\sqrt{3} + i)^2 = 4(\cos \pi/3 + i \sin \pi/3)$$

Product of Complex numbers

Definition 6

If $z = a + bi$ and $w = c + di$, the product

$$\begin{aligned} zw &= (a + bi)(c + di) \\ &= ac + adi + bci + bdi^2 \\ &= ac + (ad + bc)i - bd, \quad i^2 = -1 \\ &= (ac - bd) + (ad + bc)i \end{aligned}$$

Product of Complex numbers CONT'D

Example 7

Find the product of $z = 2 - i$ and $w = 3 - 4i$

Solution 8

$$\begin{aligned}zw &= (2 - i)(3 - 4i) \\&= 6 - 8i - 3i + 4i^2 \\&= 6 - 11i - 4, \quad i^2 = -1 \\&= 6 - 4 - 11i \\&= 2 - 11i\end{aligned}$$

- Always remember to put the real part of a complex number first.

Example 9

Given that $z = r_1(\cos \theta + i \sin \theta)$ and $w = r_2(\cos \phi + i \sin \phi)$, Find zw .

$$\begin{aligned} zw &= r_1(\cos \theta + i \sin \theta) \cdot r_2(\cos \phi + i \sin \phi) \\ &= r_1 r_2 (\cos \theta + i \sin \theta)(\cos \phi + i \sin \phi) \\ &= r_1 r_2 (\cos \theta \cos \phi + i \cos \theta \sin \phi + i \sin \theta \cos \phi + i^2 \sin \theta \sin \phi) \\ &= r_1 r_2 (\cos \theta \cos \phi - \sin \theta \sin \phi) + i(\cos \theta \sin \phi + \sin \theta \cos \phi) \end{aligned}$$

But by Trig identity:

$$\begin{aligned} \cos(\theta + \phi) &= \cos \theta \cos \phi - \sin \theta \sin \phi \\ \sin(\theta + \phi) &= \cos \theta \sin \phi + \sin \theta \cos \phi \end{aligned}$$

$$zw = r_1 r_2 (\cos(\theta + \phi) + i \sin(\theta + \phi)).$$

Product of Complex numbers in polar form

Example 10

Find the product of the numbers $z = 1 + i$ and $w = 1 + \sqrt{3}i$ using polar form of complex numbers.

Product of Complex numbers in polar form

Solution 11

$$\text{For } z = 1 + i_1 \quad x = 1, \quad y = 1$$

$$r = \sqrt{1+1} = \sqrt{2}$$

$$\theta = \tan^{-1}(1) = \pi/4$$

$$\Rightarrow z = \sqrt{2}(\cos \pi/4 + i \sin \pi/4)$$

$$\text{For } w = 1 + \sqrt{3}i, \quad x = 1, \quad y = \sqrt{3}$$

$$r^2 = 1^2 + (\sqrt{3})^2$$

$$= 1 + 3 = 4$$

$$\Rightarrow r = \sqrt{4} = 2$$

$$q = \tan^{-1}(\sqrt{3}/1)$$

$$= \tan^{-1}(\sqrt{3}) = \pi/3$$

$$\Rightarrow w = 2(\cos \pi/3 + i \sin \pi/3)$$

solution CONT'D

$$\text{Now } zw = \sqrt{2}(\cos \pi/4 + i \sin \pi/4)2 \left(\cos \pi/3 + i \sin \frac{\pi}{3} \right)$$

$$\begin{aligned} zw &= 2\sqrt{2}(\cos(\pi/4 + \pi/3) + i \sin(\pi/4 + \pi/3)) \\ &= 2\sqrt{2}(\cos 7\pi/12 + i \sin 7\pi/12) \end{aligned}$$

Division of complex numbers

Definition 12

If $z = a + bi$ and $w = c + di$ are complex numbers, then the division of complex numbers is given by $\frac{z}{w} = \frac{a+bi}{c+di}$, where we multiply the denominator $c+di$ by its conjugate $c-di$ to get rid of the complex number in the denominator.

Example 13

Simplify $\frac{8-3i}{6i}$

Division of complex numbers

Solution 14

$$\begin{aligned}\frac{8 - 3i}{6i} &= \frac{(8 - 3i) 6i}{6i \cdot 6i} \\&= \frac{48i - 18i^2}{36i^2} \\&= \frac{48i - 18(-1)}{36(-1)} \\&= \frac{18 + 48i}{-36} \\&= -\frac{1}{2} - \frac{4}{3}i\end{aligned}$$

Division example CONT'D

Example 15

If $z = r_1(\cos \theta + i \sin \theta)$ and $w = r_2(\cos \phi + i \sin \phi)$ are complex numbers find the quotient of z and w .

Division example CONT'D

Solution 16

$$\begin{aligned}\frac{z}{w} &= \frac{r_1(\cos \theta + i \sin \theta)}{r_2(\cos \phi + i \sin \phi)} \\&= \frac{r_1 (\cos \theta + i \sin \theta)(\cos \phi - i \sin \phi)}{r_2 (\cos \phi + i \sin \phi)(\cos \phi - i \sin \phi)} \\&= \frac{r_1 \cos \theta \cos \phi - i \sin \phi \cos \theta + i \sin \theta \cos \phi - i^2 \sin \theta \sin \phi}{r_2 \cos^2 \phi - i \cos \phi \sin \phi + i \cos \phi \sin \phi - i^2 \sin^2 \phi} \\&= \frac{r_1 \cos \theta \cos \phi - i \sin \phi \cos \theta + i \sin \theta \cos \phi + \sin \theta \sin \phi}{r_2 \cos^2 \phi + \sin^2 \phi} \\&= \frac{r_1 \cos \theta \cos \phi + \sin \phi \sin \theta + i(\sin \theta \cos \phi - \sin \phi \cos \theta)}{r_2 \cdot 1} \\&= \frac{r_1}{r_2} (\cos(\theta - \phi) + i \sin(\theta - \phi))\end{aligned}$$

THE END